

Problem 7. In class, we saw four ways of representing linear subspaces of $\mathbb{R}^n$:

1. The span of a collection of vectors: $\text{span}\{v_1, \dots, v_r\}$.
2. The image of a matrix: $\text{im}(A)$.
3. The solution set of a homogeneous linear system:

$$\text{the set of all vectors } \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \text{ such that } \begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = 0 \\ \vdots \\ a_{r1}x_1 + \dots + a_{rn}x_n = 0 \end{cases}$$

4. The kernel of a matrix: $\ker(A)$, the set of all vectors v such that $Av = 0$.

In each part, you are given a representation of a subspace in one of these forms. Find the the three other representations of it.

(a)

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

(b)

$$\ker \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}$$

(c) The solution set of the system of equations

$$\begin{cases} x_1 - 2x_2 - x_3 + 5x_4 = 0 \\ 3x_1 + x_2 + x_3 + 3x_4 = 0 \end{cases}$$

(d)

$$\text{im} \begin{bmatrix} 1 & -1 & 2 \\ 3 & -3 & 6 \\ 1 & -1 & 2 \end{bmatrix}$$

Solution.

(a) We are given the subspace in the form of a span:

$$U = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

To express it as the image of a matrix, we use the vectors as columns:

$$U = \text{im} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Now we want to defining equations for this subspace. A vector

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

is in U if and only if the linear system with augmented matrix

$$\left[\begin{array}{cc|c} 1 & 1 & x_1 \\ 1 & 2 & x_2 \\ 1 & 0 & x_3 \\ -1 & 1 & x_4 \end{array} \right]$$

is solvable. Applying row operations, this reduces as follows:

$$\left[\begin{array}{cc|c} 1 & 1 & x_1 \\ 1 & 2 & x_2 \\ 1 & 0 & x_3 \\ -1 & 1 & x_4 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 1 & x_1 \\ 0 & 1 & x_2 - x_1 \\ 0 & -1 & x_3 - x_1 \\ 0 & 2 & x_4 + x_1 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 2x_1 - x_2 \\ 0 & 1 & x_2 - x_1 \\ 0 & 0 & x_3 + x_2 - 2x_1 \\ 0 & 0 & x_4 + 3x_1 - 2x_2 \end{array} \right]$$

The condition for the system to be solvable is that the entries in the last two rows are zero, that is:

$$\begin{cases} -2x_1 + x_2 + x_3 = 0 \\ 3x_1 - 2x_2 + x_4 = 0 \end{cases}$$

U is the solution set of this system of equations. Putting the coefficients of these equations into the rows of a matrix, we have:

$$U = \ker \begin{bmatrix} -2 & 1 & 1 & 0 \\ 3 & -2 & 0 & 1 \end{bmatrix}$$

(b) We are given the subspace in the form of a kernel:

$$V = \ker \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}$$

This is equivalent to the system of equations

$$\begin{cases} x_1 + x_2 + x_3 = 0 \\ x_1 + 2x_2 + 3x_3 = 0 \\ x_1 + 3x_2 + 5x_3 = 0 \end{cases}$$

To find a spanning set for V , we solve this system:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 0 \\ 1 & 3 & 5 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & 4 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

This corresponds to equations

$$\begin{cases} x_1 - x_3 = 0 \\ x_2 + 2x_3 = 0 \end{cases}$$

Our free variable is x_3 , and we have $x_1 = x_3$, $x_2 = -2x_3$. Thus a general solution has the form

$$\begin{bmatrix} x_3 \\ -2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

and so

$$V = \text{span} \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\} = \text{im} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

(c) We are given our subspace W as the solution set of the linear system:

$$\begin{cases} x_1 - 2x_2 - x_3 + 5x_4 = 0 \\ 3x_1 + x_2 + x_3 + 3x_4 = 0 \end{cases}$$

Putting the coefficients of the linear system into the rows of a matrix, we have:

$$W = \ker \begin{bmatrix} 1 & -2 & -1 & 5 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

We solve the linear system as follows

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 5 & 0 \\ 3 & 1 & 1 & 3 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cccc|c} 1 & -2 & -1 & 5 & 0 \\ 0 & 7 & 4 & -12 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cccc|c} 1 & -2 & -1 & 5 & 0 \\ 0 & 1 & 4/7 & -12/7 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cccc|c} 1 & 0 & 1/7 & 11/7 & 0 \\ 0 & 1 & 4/7 & -12/7 & 0 \end{array} \right]$$

Our free variables are x_3 and x_4 , and we have

$$\begin{aligned} x_1 &= -\frac{x_3}{7} - \frac{11x_4}{7} \\ x_2 &= -\frac{4x_3}{7} + \frac{12x_4}{7} \end{aligned}$$

Thus our general solution is

$$\begin{bmatrix} -x_3/7 - 11x_4/7 \\ -4x_3/7 + 12x_4/7 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -1/7 \\ -4/7 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -11/7 \\ 12/7 \\ 0 \\ 1 \end{bmatrix}$$

We can write W as a span as follows:

$$W = \text{span} \left\{ \begin{bmatrix} -1/7 \\ -4/7 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -11/7 \\ 12/7 \\ 0 \\ 1 \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} -1 \\ -4 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} -11 \\ 12 \\ 0 \\ 7 \end{bmatrix} \right\}$$

and as an image as follows:

$$W = \text{im} \begin{bmatrix} 1/7 & -11/7 \\ -4/7 & 12/7 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \text{im} \begin{bmatrix} -1 & -11 \\ -4 & 12 \\ 7 & 0 \\ 0 & 7 \end{bmatrix}$$

For both the span and image representations, both the first and second form work, the second just multiplies through by 7 to get rid of the denominators.

(d) Here we are given the subspace as an image:

$$X = \text{im} \begin{bmatrix} 1 & -1 & 2 \\ 3 & -3 & 6 \\ 1 & -1 & 2 \end{bmatrix}$$

This is equivalent to the span representation

$$X = \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \right\}$$

We only need the first vector, because the other two are scalar multiples of it, but writing the linear subspace as the span of all three is still correct.

To find defining equations for this subspace, we use the same process as in part (a); we want to find conditions for when the system with augmented matrix

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & x_1 \\ 3 & -3 & 6 & x_2 \\ 1 & -1 & 2 & x_3 \end{array} \right]$$

is solvable. The row operations go as follows:

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & x_1 \\ 3 & -3 & 6 & x_2 \\ 1 & -1 & 2 & x_3 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & -1 & 2 & x_1 \\ 0 & 0 & 0 & x_2 - 3x_1 \\ 0 & 0 & 0 & x_3 - x_1 \end{array} \right]$$

The defining equations are then

$$\begin{cases} -3x_1 + x_2 = 0 \\ -x_1 + x_3 = 0 \end{cases}$$

This is equivalent to the kernel representation

$$X = \ker \begin{bmatrix} -3 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

In this case, there are other ways to find the defining equations. Recognizing that our space was really the span of a single vector, we could have also considered the linear system with augmented matrix

$$\left[\begin{array}{c|c} 1 & x_1 \\ 3 & x_2 \\ 1 & x_3 \end{array} \right]$$

Alternatively, we can use the following simple argument: If

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = t \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

then $x_1 = t = x_3$ and $x_2 = 3t = 3x_1$. Therefore $-3x_1 + x_2 = 0$ and $-x_1 + x_3 = 0$. Conversely, if these equations are satisfied, then we have

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \in \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \right\}$$